RESEARCH REPORT

Multi-Strategy Framework & Regime Detection Upgrade

Comprehensive Project Report — Combined Edition

IIT Kanpur | Finance & Analytics Club

Project: IIT-K Options Greeks PS Framework

Part I: Multi-Strategy Backtester (v1 $\rightarrow$ v2) | Part II: HMM/GMM Regime Detection Upgrade

Backtest window: 1 Jan 2022 – 31 Dec 2024

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PART I

NIFTY Options Backtester — Multi-Strategy Framework

1 Executive Summary

This report documents the complete development lifecycle of a NIFTY 50 options backtesting framework built for the IIT Kanpur Finance & Analytics Club. The project evolved across two major iterations — from a focused two-strategy comparison tool to a production-grade multi-strategy backtester — all implemented entirely in Python using freely available data from Yahoo Finance and a self-contained Black-Scholes pricing engine.

The framework backtests options strategies on NIFTY 50 over the 2022–2024 period (three calendar years, $\sim$740 trading days) using a simulated Implied Volatility surface derived from rolling Historical Volatility, making it fully self-sufficient without requiring paid data subscriptions.

Key Outcomes from the Final Multi-Strategy Run

- 67 total trades executed across 4 strategies over the 2022–2024 backtest window.
- Net P&L of ₹26,572 (+5.31% return) on ₹5,00,000 initial capital.
- Overall win rate of 62.7% with a profit factor of 1.27.
- Maximum portfolio drawdown controlled at -8.22%.
- Star performer: Bull Put Spread on Dips — ₹36,238 profit (+7.25%), 84.6% win rate.
- Two strategies were net positive; two required further calibration.

2 Project Context & Motivation

The project was assigned as part of the IIT Kanpur Finance & Analytics Club's Options Greeks Problem Statement (PS) Framework. The mandate was to design, implement, and backtest systematic options strategies on Indian equity derivatives (NIFTY 50), with a specific focus on using the five Greeks — Delta, Gamma, Vega, Theta, and Rho — as entry, monitoring, and exit signals.

2.1 Design Constraints & Objectives

- 100% free infrastructure — no paid data APIs (NSE Options chain, Bloomberg, Refinitiv).
- All pricing must use Black-Scholes with a simulated IV surface built from realized volatility.
- Strategies must be systematic and rule-based, with all trade logic documented.

- Greek thresholds (Delta alerts, Gamma exits, Vega exposure limits) must be explicitly managed.
- The backtester must generate machine-readable results (JSON) and visual dashboards (HTML).
- Designed for Google Colab compatibility — minimal dependencies, single-file execution.

3 System Architecture

The system is structured into eight loosely-coupled sections across two major versioned files. The architecture was designed to be modular — each strategy is implemented as an independent class with its own entry logic, daily monitoring loop, and exit conditions. A shared Black-Scholes engine and a single data-fetch layer serve all strategies.

3.1 Version History

Version	File	Description	Strategies
v1.0	`greeks_backtester.py` (original)	First backtester — Iron Condor + Long Straddle two-way comparison	2
v1.1	`Backtester.ipynb`	Notebook runner wrapping v1 — adds `LongStraddleBacktester` class and side-by-side HTML dashboard	2
v2.0	`greeks_backtester__1_.py`	Full multi-strategy rewrite — 4–6 strategy classes, trend indicators, SMA filter, dip detection, combined portfolio equity curve	4 (active)

Table 1: Version history of the backtester codebase.

3.2 Module Breakdown

Section	Contents
Section 1 — Configuration	Single `CFG` dict with all tunable parameters: dates, capital, per-strategy offsets, Greek thresholds, entry/exit rules
Section 2 — Black-Scholes Engine	`_bs_price()` raw pricer + `greeks()` returning {price, delta, gamma, vega, theta, iv} per option leg
Section 3 — Data Layer	`fetch_nifty()`: yfinance download + HV10/HV20/HV30 + IV estimate + IV/HV ratio + SMA20/50/100 + trend + 5d return
Section 4 — Strategy Backtester Classes	`IronCondorBacktester`, `CreditPutSpreadBacktester`, `LongStraddleBacktester`, `BullPutSpreadDipBacktester`
Section 5 — Metrics Engine	`compute_metrics()`, `compute_strategy_metrics()`: Sharpe, drawdown, profit factor, win rate, exit breakdown
Section 6 — Visualization	`plot_multi_strategy_dashboard()`: Plotly dark-theme HTML dashboard (portfolio curve, drawdown, per-trade bars, strategy pie)
Section 7 — Print Reports	`print_combined_summary()`, `print_trade_log()`: console-formatted tabular output
Section 8 — Entry Point	`run_all_strategies()`: orchestrates all 4 strategies and saves `multi_strategy_results.json`

Table 2: Module-by-module breakdown of the v2.0 codebase.

4 Data Pipeline & IV Surface Construction

Since live NSE options chain data requires paid subscriptions, the project constructs a synthetic Implied Volatility surface from realized (historical) volatility using rolling standard deviation of log returns. This is a standard industry approximation valid for research and strategy development purposes.

4.1 Data Source

- Ticker: `^NSEI` (NIFTY 50 Index) via `yfinance.download()`.
- Fields used: Close price (auto-adjusted) for all Greek and volatility calculations.
- Date range: 2022-01-01 to 2024-12-31 (3 years, ∼740 trading days).
- MultiIndex flattening applied for compatibility with newer yfinance versions.

4.2 Volatility Surface Construction

Signal	Formula	Purpose
HV10	10-day rolling std(log returns) $\times\sqrt{252}$	Short-term noise; vol spike detection
HV20	20-day rolling std(log returns) $\times\sqrt{252}$	Medium-term baseline; primary IV input
HV30	30-day rolling std(log returns) $\times\sqrt{252}$	Slow-moving regime baseline for IV/HV ratio
IV	HV20 + 3% (clipped to 10%–80%)	Implied Volatility estimate used in Black-Scholes pricing
IV_HV_ratio	IV / HV30	Regime signal: >1.10 = elevated vol (sell), <1.00 = depressed (buy)
vol_gap	IV $-$ HV10	Selling edge: positive means IV above short-term realized vol
SMA20/50/100	Rolling close means at 20/50/100 days	Trend filter for directional strategies
trend	1 if SMA50 $>$ SMA100 else -1	Binary trend label — bullish or bearish regime
ret_5d	pct_change(5)	Recent momentum; range-bound filter for Iron Condor entry

Table 3: Volatility and trend signals computed from raw price data.

4.3 Expiry Calendar Generation

NIFTY monthly expiries fall on the last Thursday of each month. The `get_monthly_expiry_dates()` function programmatically generates these dates by walking backward from each month's last day to the nearest Thursday, then adjusting for market holidays (if the Thursday is not a trading day, it steps back to the previous trading day). This produces 36 valid expiry dates over the 2022–2024 window.

5 Options Strategies — Detailed Specification

5.1 Iron Condor (Range-Bound Market)

Rationale: Earn theta decay in low-volatility, range-bound regimes by collecting premium on both sides. Defined-risk due to long wings.

Parameter	v1 Setting	v2 Setting (Optimized)	Rationale
Leg structure	4-leg: sell/buy call spread + sell/buy put spread	Same	Defined max loss via wings
Entry filter	IV/HV $>$ 1.10	IV/HV $>$ 1.20 + |5d move| $<$ 2.5%	Stricter — only truly elevated vol + calm market
Strike offsets	CALL/PUT $\pm 5 \times 50 = \pm 250$pt from ATM	$\pm 8 \times 50 = \pm 400$pt from ATM (wider!)	More room for spot movement
Wing width	$3 \times 50 = 150$pt protection	$3 \times 50 = 150$pt (same)	Consistent defined-risk buffer
Profit target	50% of premium collected	35% — faster exit	Reduce tail risk exposure
Stop loss	2.0$\times$ net premium	1.3$\times$ — very tight	Cut losses fast in trending moves
Time exit	4 DTE	7 DTE	Exit earlier to avoid gamma risk near expiry
Net credit formula	(sell_call $-$ buy_call) + (sell_put $-$ buy_put)	Same	—

Table 4: Iron Condor parameters, v1 vs. v2.

5.2 Credit Put Spread (Bullish Trend)

Rationale: Earn premium in bullish markets by selling an OTM put and buying a further OTM put. Profit if NIFTY stays above the short put strike.

Parameter	Value
Entry filter	IV/HV $>$ 1.10 AND spot $>$ SMA100 $\times$ 1.02 (2% above trend)
Strike structure	Sell put at ATM -6×50 (300pt below); Buy put at ATM -9×50 (450pt below)
Net credit	(short put price) $-$ (long put price)
Profit target	45% of net credit
Stop loss	2.0$\times$ net premium paid to close
Time exit	5 DTE

Table 5: Credit Put Spread parameters.

5.3 Long Straddle (Low IV / Volatility Expansion)

Rationale: Buy ATM call + ATM put when IV is depressed. Profit from a large directional move in either direction (earnings-type events, breakouts).

Parameter	v1 Setting	v2 Setting (Improved)	Rationale
Entry filter	IV/HV $<$ 1.00	IV/HV $<$ 1.18 (relaxed)	Increase trade frequency; still below fair value
Profit target	50% gain on combined position	Same	Let winners run
Stop loss	25% loss on combined position	18% — tighter	Theta accelerates; cut faster
IV crush exit	Exit if IV drops 12%	Same	Protect against vol normalization
Max hold	15 days	10 days	Shorter theta exposure window
Leg-lock logic	N/A	NEW: lock profitable leg as ITM, let other run	Capture directional momentum

Table 6: Long Straddle parameters, v1 vs. v2.

5.4 Bull Put Spread on Dips (Mean-Reversion) — Star Strategy

Rationale: On market dips (5-day decline $\geq$ 2.5%), sell an OTM put spread — expecting mean-reversion. NIFTY has historically recovered from moderate pullbacks, making this strategy naturally aligned with market behavior.

Parameter	Value
Entry trigger	5-day return ≤ -2.5% (meaningful dip)
Trend requirement	SMA50 $>$ SMA100 (overall bullish regime)
Entry DTE	15 DTE (shorter hold than other strategies)
Strike structure	Sell put at ATM -4×50 (200pt below); Buy put at ATM -7×50 (350pt below)
Profit target	45% of net credit
Stop loss	2.0$\times$ net premium — wide stop to allow mean-reversion time
Cooldown	7 days after each trade — prevent over-trading on sustained trends
Time exit	3 DTE

Table 7: Bull Put Spread on Dips parameters.

6 Greek Management Framework

All strategies share a common Greek monitoring framework, applied daily during each trade's hold period. Greek thresholds generate alert and hard-exit signals, and are computed fresh each day using the current spot price, current simulated IV, and remaining DTE.

6.1 Black-Scholes Formulae Implemented

All Greeks are computed analytically (closed-form). Inputs: S (spot), K (strike), T (time to expiry in years), r (risk-free rate = 6.5%), σ (IV). Key formulae:

Greek	Formula	Interpretation
Delta (Δ)	$N(d_1)$ for CE; $N(d_1) - 1$ for PE	Rate of change of option price w.r.t. spot
Gamma (Γ)	$N'(d_1)/(S\sigma\sqrt{T})$	Rate of change of Delta; convexity risk
Vega (ν)	$S\,N'(d_1)\sqrt{T}/100$	Sensitivity to 1% change in IV
Theta (θ)	$-[SN'(d_1)\sigma/(2\sqrt{T})]/365 \pm rKe^{-rT}N(d_2)$	Daily time decay (per calendar day)
d_1	$(\ln(S/K) + (r + \sigma^2/2)T)/(\sigma\sqrt{T})$	Standardized moneyness measure
d_2	$d_1 - \sigma\sqrt{T}$	Risk-neutral probability ITM driver

Table 8: Black-Scholes Greek formulae used by the pricing engine.

6.2 Net Greek Computation for Multi-Leg Strategies

For spread strategies, net Greeks are computed by summing across all four legs with correct signs. For the Iron Condor (short the outer, long the wing):

$$\begin{aligned}
\text{Net Delta} &= -(\Delta_{\text{short call}} - \Delta_{\text{long call}}) - (\Delta_{\text{short put}} - \Delta_{\text{long put}}) \\
\text{Net Gamma} &= -(\Gamma_{\text{short call}} - \Gamma_{\text{long call}}) - (\Gamma_{\text{short put}} - \Gamma_{\text{long put}}) \\
\text{Net Vega} &= -(\nu_{\text{short call}} - \nu_{\text{long call}}) - (\nu_{\text{short put}} - \nu_{\text{long put}}) \\
\text{Net Theta} &= -(\theta_{\text{short call}} - \theta_{\text{long call}}) - (\theta_{\text{short put}} - \theta_{\text{long put}}) \quad \text{[positive for sellers]}
\end{aligned}$$

6.3 Shared Alert & Exit Thresholds

Greek	Alert	Hard Exit	Action on Alert	Action on Exit
Delta (|net Δ|)	> 0.25	> 0.40	Log alert, increase monitoring	Immediate position close
Gamma (|net Γ|)	> 0.0015	> 0.0020	Log alert	Immediate position close
Vega (IV rise from entry)	$> +25\%$ relative	$> +45\%$ relative	Log alert	Immediate position close

Table 9: Greek-based alert and hard-exit thresholds shared across all strategies.

7 Backtest Results & Performance Analysis

Backtest Period: 2022-01-01 to 2024-12-31 | **Initial Capital:** ₹5,00,000 | **Lot Size:** 50 units

7.1 Combined Portfolio Performance

Metric	Value
Total Trades (closed)	67
Net P&L	₹26,572
Total Return	$+5.31$%
Win Rate	62.7%
Average Winning Trade	₹2,967
Average Losing Trade	−₹3,922
Best Single Trade	₹8,302
Worst Single Trade	−₹7,692
Profit Factor	1.27
Maximum Drawdown	-8.22%

Table 10: Combined portfolio performance, 2022–2024 backtest.

7.2 Per-Strategy Performance Breakdown

Strategy	P&L (₹)	Return %	Trades	Win Rate	Profit Factor	Max DD %
Bull Put Spread (Dip) $\star$	**+36,238**	+7.25%	26	**84.6%**	2.52	-2.63%
Long Straddle	+20,123	+4.02%	4	100.0%	∞ (999)	0.00%
Iron Condor	−**16,806**	-3.36%	22	36.4%	0.61	-4.75%
Credit Put Spread	−**12,983**	-2.60%	15	53.3%	0.59	-2.58%

Table 11: Per-strategy performance breakdown ($\star$ = star performer).

7.3 Exit Reason Analysis

Understanding how trades exit reveals strategy behavior and risk characteristics:

Exit Reason	Count	Interpretation
PROFIT_TARGET	35 (52.2%)	Majority of trades hit the profit threshold — strategies are correctly calibrated
STOP_LOSS	24 (35.8%)	Significant stop-loss rate, especially in IC and CPS — signals regime mismatch
TIME_EXIT	4 (6.0%)	DTE-based expiry exit — normal end-of-cycle behavior
LEG_LOCK_PROFIT	4 (6.0%)	Long Straddle leg-locking logic — directional winner captured while other leg runs

Table 12: Exit reason distribution across all 67 trades.

7.4 Key Observations

- Bull Put Spread on Dips dominates, with 26 trades and 84.6% win rate. The NIFTY's structural upward bias means dip-and-recover patterns are frequent — this strategy is naturally aligned with Indian market behavior.
- Long Straddle delivered a 100% win rate on 4 trades using the leg-lock mechanism. The small trade count suggests the IV filter was still selective even after relaxation — a larger sample is needed for robust conclusions.
- Iron Condor underperformed due to the 2022–2024 period containing multiple trending episodes (Russia–Ukraine, FII selloff, US rate hike cycle) that blew through the wings despite the wider 400pt offset in v2.
- Credit Put Spread faced loss-asymmetry: average loss (₹4,480) exceeded average win (₹2,297) — the 2.0× stop multiplier and 300pt offset still left significant tail exposure in sharp down-moves.

8 Version Evolution: v1 vs. v2 Comparison

The project underwent a major architectural and strategy overhaul between v1 and v2.

8.1 v1.0 — Dual Strategy Baseline (`Backtester.ipynb`)

- Two strategies: Iron Condor (vol selling) vs. Long Straddle (vol buying) — a direct comparison framework.
- Single `IronCondorBacktester` and `LongStraddleBacktester` class — each runs its own equity curve and PV log.
- Side-by-side Plotly dashboard: 3×2 grid — portfolio value, drawdown, per-trade bars for each strategy.
- Iron Condor parameters: $\pm 5 \times 50$ offsets, 50% profit target, 2.0× stop loss, 4 DTE exit.
- Long Straddle: IV/HV $<$ 1.00 entry filter (strict), 25% stop loss, 15-day max hold.
- No trend indicators — regime-blind for directional strategies.

8.2 v2.0 — Multi-Strategy Production Build (`greeks_backtester__1_.py`)

- Expanded to 4–6 strategy classes; 4 run in the active configuration.
- New trend infrastructure: SMA20/50/100, trend signal, 5-day return for momentum/dip detection.
- Iron Condor v2: wider strikes (400pt), stricter IV filter (1.20), range-bound filter (|5d| < 2.5%), faster exits.
- Credit Put Spread: NEW — bullish-biased spread with SMA100 trend confirmation.
- Long Straddle v2: relaxed IV filter ($<$1.18), leg-lock profit mechanism, tighter 18% stop, 10-day cap.
- Bull Put Spread on Dips: NEW star strategy — dip threshold, 7-day cooldown, mean-reversion oriented.
- Dropped Short Strangle (undefined loss risk) and Credit Call Spread (bearish bias in structural bull market).
- Portfolio-level metrics: combined equity curve, overall drawdown, cross-strategy P&L attribution pie.
- Unified trade log with strategy labels — all 67 trades sortable and comparable in a single table.

8.3 Parameter Optimization Summary

Parameter	v1	v2	Direction	Rationale
IC Strike offset	$5 \times 50 = 250$pt	$8 \times 50 = 400$pt	Wider	Give more room before wings are breached
IC IV/HV min	1.10	1.20	Stricter	Trade only when vol is significantly elevated
IC Profit target	50%	35%	Lower	Exit faster; reduce holding risk
IC Stop loss	2.0×	1.3×	Tighter	Cap per-trade loss more aggressively
IC Time exit	4 DTE	7 DTE	Earlier	Avoid gamma explosion near expiry
LS IV/HV max	1.00	1.18	Relaxed	Increase trade frequency
LS Stop loss	25%	18%	Tighter	Theta accelerates — cut losses faster
LS Max hold	15 days	10 days	Shorter	Reduce theta bleed exposure
Capital risk/trade	12%	8%	Lower	Better position sizing; drawdown control
Max lots/trade	3	2	Lower	Reduce concentration risk per trade

Table 13: Parameter changes from v1 to v2 across strategies (IC = Iron Condor, LS = Long Straddle).

9 Visualization & Output Artifacts

The backtester generates three types of outputs: live console logs, machine-readable JSON results, and interactive HTML dashboards.

9.1 Dashboard: `multi_strategy_dashboard.html`

A dark-themed Plotly interactive HTML dashboard with the following panels:

- **Panel 1** (top-left): Combined portfolio equity curve — daily portfolio value from ₹5L initial capital, colored by strategy contribution, fill-to-zero gradient.
- **Panel 2** (top-right): Per-trade P&L by strategy — stacked bar chart with strategy color coding; green = profit, red = loss.
- **Panel 3** (bottom-left): Combined portfolio drawdown — red fill-to-zero; maximum drawdown line annotated.

- **Panel 4** (bottom-right): Strategy P&L pie chart — proportional contribution of each strategy to total P&L.
- **Title banner**: live performance stats (trades, win rate, P&L, return, profit factor, max DD).

Theme: `#0f172a` background, `#1e293b` plot area, JetBrains Mono font — consistent with quantitative finance dashboard standards.

9.2 Dashboard: `comparison_dashboard.html` (v1)

The v1 comparison dashboard uses a 3×2 subplot grid for Iron Condor vs. Long Straddle:

- Row 1: Portfolio value over time (area chart with gradient fill).
- Row 2: Drawdown series (red area).
- Row 3: Per-trade P&L bar chart (green/red by profitability).
- Strategy stats summarized in title subtitle — trades, win rate, P&L, return, Sharpe, max drawdown.

9.3 JSON Result Files

File	Contents
`multi_strategy_results.json`	`combined_metrics` (dict), `strategy_metrics` (dict per strategy), `trades` (list of 67 trade dicts with full entry/exit details)
`comparison_results.json` (v1)	`iron_condor` and `long_straddle` sub-dicts, each containing metrics and trades lists
`backtest_results_ic.json`	Iron Condor only — metrics, trades, alerts, `pv_log` (daily portfolio value series)
`backtest_results_ls.json`	Long Straddle only — same structure

Table 14: Machine-readable JSON output files produced by the backtester.

9.4 Trade Record Schema

Every trade is stored as a dict with the following fields:

Field	Description
id	Sequential trade ID across all strategies
strategy	Strategy name (Iron Condor / Credit Put Spread / Long Straddle / Bull Put Spread (Dip))
entry_date / exit_date	YYYY-MM-DD strings
entry_spot / exit_spot	NIFTY 50 closing price on entry and exit days
entry_iv / exit_iv	Simulated IV on each day
K_short_call, K_long_call, K_short_put, K_long_put	Strike prices for all four legs (where applicable)
entry_net_premium	Net credit/debit received at entry (per unit)
lots	Number of lots traded (lot size = 50)
exit_reason	PROFIT_TARGET / STOP_LOSS / TIME_EXIT / LEG_LOCK_PROFIT / OPEN_AT_END
pnl	Realized P&L in ₹ for this trade
dte_held	Number of trading days from entry to exit
entry_delta, entry_gamma, entry_vega, entry_theta	Net Greeks at entry

Table 15: Schema for each trade record in the JSON output.

10 Technical Stack & Dependencies

The entire project runs on a Python-only stack with no paid data APIs. The key dependency choices were made for Colab compatibility and zero-cost deployment.

Library	Version	Usage
yfinance	latest	Download NIFTY 50 historical OHLCV data (^NSEI)
pandas	$\geq$ 1.5	Time series manipulation, rolling calculations, trade log DataFrames
numpy	$\geq$ 1.22	Log return computation, vectorized HV calculations, BS formula math
scipy.stats.norm	from scipy	$N(d_1)$, $N(d_2)$ for Black-Scholes call/put pricing and Greeks
plotly.graph_objects	$\geq$ 5.0	Interactive dark-theme dashboards; `fig.write_html()` output
plotly.subplots	$\geq$ 5.0	`make_subplots` for multi-panel dashboard layout
json	stdlib	Serialization of all trade and metric results
warnings	stdlib	Suppress pandas `FutureWarning` noise during backtest

Table 16: Python dependencies used by the backtester.

Google Colab Installation Command:

```
!pip install yfinance pandas numpy scipy plotly -q
```

11 Known Limitations & Suggested Improvements

11.1 Limitations of the Current Framework

- **Synthetic IV**: The IV surface is approximated as HV20 + 3%. Real NSE options have a volatility smile and skew not captured by this flat surface. This understates the cost of OTM options.
- **No bid-ask spread**: Entry and exit prices assume mid-market. Real NIFTY options have bid-ask spreads of ₹2–10 per unit for liquid strikes — this can significantly erode credit spread profits.
- **No slippage or transaction costs**: NSE charges, STT, and broker commissions are not deducted. For 67 trades with 4 legs each, transaction costs could be ₹5,000–15,000.
- **Monthly entry only**: The backtester enters once per monthly expiry cycle. Weekly NIFTY options (available since 2019) would provide $\sim 4\times$ more trading opportunities.
- **IV is non-stochastic**: The IV model does not allow for regime shifts, volatility jumps, or mean-reversion of vol — stochastic vol models (Heston, SABR) would be more realistic.
- **Small sample for Long Straddle (4 trades)**: The 100% win rate is statistically unreliable. A longer backtest or relaxed filters are needed for robust conclusions.

11.2 Recommended Next Steps

- Add a transaction cost layer: ₹20/trade brokerage + NSE charges (STT = 0.05% of premium for options).
- Move to weekly expiry: use NSE Thursday weekly expiries for higher trade frequency and better theta management.
- Real NSE data integration: NSEpy or an NSE option chain scraper for actual market IV instead of synthetic.
- Walk-forward validation: split 2022–2024 into in-sample calibration (2022–2023) and out-of-sample test (2024).
- Portfolio margin netting: in a real multi-strategy portfolio, margin offsets exist — implement margin calculations.
- Add RSI/momentum filters to the dip strategy: filter false dips (sustained down-trends) vs. true mean-reversion setups.
- Sharpe ratio and Sortino ratio: add risk-adjusted return metrics to the metrics engine.
- Monte Carlo simulation: run IV bootstrapping to stress-test strategy P&L under different volatility regimes.

Forward link: a subsequent iteration of this project added HMM/GMM-based market-regime detection on top of this v2.0 engine, gating each strategy to trade only in its best-suited regime. That upgrade is documented separately in the companion *Regime Detection Upgrade* report and improved total P&L by 47%, profit factor by 84%, and cut maximum drawdown by 77% relative to the results shown here.

Appendix: Sample Trades from Each Strategy

Representative trades illustrating each strategy's behavior:

Iron Condor — Sample Trade

Field	Value
Trade ID	#1
Entry Date	2022-02-15 | Expiry: 2022-03-31
NIFTY Spot	₹17,352
Structure	Sell 17,750 Call / Buy 17,900 Call | Sell 16,950 Put / Buy 16,800 Put
Net Premium	₹107.88 per unit × 2 lots × 50 = ₹10,788 gross
Entry IV	26.92%
Exit	2022-03-23 via TIME_EXIT (at 7 DTE)
P&L	**+₹2,785**

Table 17: Sample Iron Condor trade record.

Bull Put Spread on Dips — Typical Trade Profile

This strategy typically entered on sharp 1-week NIFTY declines (2.5%+), placed a put spread ~200pt below ATM, and exited within 15 days at 45% profit. The 2022–2024 period had multiple such dip-and-recover events (budget reactions, FII selling episodes, global macro events) that created favorable entry conditions. With 26 trades and 21 profit-target exits, the strategy demonstrated a high hit-rate and disciplined sizing.

Long Straddle — Leg-Lock Mechanism Example

The improved Long Straddle uses a "leg-lock" exit: when one leg becomes deep ITM and the other leg has residual value, the profitable leg is closed (realizing most of the gain) while the other leg continues to run. All 4 straddle trades exited via LEG_LOCK_PROFIT, with P&Ls of ₹674, ₹2,963, ₹8,302, and ₹8,184 — confirming the mechanism works well when the move is genuine.

PART II

Regime Detection Upgrade — HMM/GMM vs. Baseline

Context: this part documents a subsequent upgrade applied on top of the v2.0 multi-strategy engine described in Part I. The "Before" / "Without HMM/GMM" column throughout this part corresponds exactly to the v2.0 results reported in Part I, Section 7.

12 Executive Summary

This report documents the integration of Hidden Markov Model (HMM) and Gaussian Mixture Model (GMM) market-regime detection into the existing NIFTY multi-strategy options backtester. The objective was to reduce the number of low-quality trades taken in the wrong market environment, without introducing any look-ahead information. Regime detection was trained using a walk-forward approach, and each of the four trading strategies was gated to fire only in the regime it is best suited for.

The upgraded engine (**With HMM/GMM**) was benchmarked against the previous version (**Without HMM/GMM**) on an identical three-year backtest. The result is a meaningfully more profitable and far less risky system: total P&L improved by 47%, the profit factor nearly doubled, and maximum drawdown fell by 77%, while the number of trades taken dropped from 67 to 43 — fewer trades, of materially higher quality.

Metric	Before (Without HMM/GMM)	After (With HMM/GMM)
Total P&L	₹26,572	**₹38,975 (+47%)**
Profit Factor	1.27	**2.34 (+84%)**
Max Drawdown	-8.22%	-1.85% **(77% less risk)**
Stop-Loss Hits	24	**1 (-96%)**

Table 18: Headline before/after comparison.

Key Result

Adding a regime-detection gate — with no change to the underlying strategy logic — improved every single headline metric simultaneously: more profit, higher quality trades, and a sharply lower drawdown.

13 What Changed

greeks_backtester.py was rewritten to add HMM/GMM-based regime detection. The core change is the addition of a regime-detection layer that sits between raw NIFTY price data and the strategy execution logic. Instead of every strategy trading on every eligible day, each strategy now only trades when the market is in the regime it statistically performs best in.

13.1 Architecture

The pipeline now flows as follows:

1. NIFTY price data is converted into features (returns, volatility, momentum, volume, IV/HV ratio).
2. A Gaussian Mixture Model clusters days into three statistical regimes; a Hidden Markov Model is layered on top to add temporal persistence so that regimes do not flicker day-to-day.
3. Regimes are re-trained on a rolling walk-forward basis (200-day training window, refit every 63 trading days), so no future information ever leaks into a trading decision.
4. Each of the four strategies is gated: it is only allowed to open a new position when the current regime matches its known profile.

Regime	Description	Share of Days
LOW_VOL_TRENDING	Calm, directional market	26%
RANGE_BOUND	Sideways / mean-reverting	40%
HIGH_VOL_STRESS	Elevated volatility / stress	34%

Table 19: The three statistical regimes discovered by the GMM/HMM layer.

Strategy-to-regime mapping used with HMM/GMM

- **Iron Condor** → trades only in `RANGE_BOUND`.
- **Credit Put Spread** → trades only in `LOW_VOL_TRENDING`.
- **Long Straddle** → trades only around the transition into `HIGH_VOL`.
- **Bull Put Spread (Dip-buy)** → skipped whenever `HIGH_VOL_STRESS` probability exceeds 60%.

13.2 Other Engine Changes

- **New `REGIME_EXIT` logic**: an open trade is closed early if the market regime shifts mid-trade, limiting damage when the market's character changes.
- **Dynamic stop-loss multipliers**: stop distance now widens in volatile regimes and tightens in calm regimes, instead of using one fixed stop for all conditions.
- **Confidence**: average regime-classification confidence across the backtest was 87.8%, indicating the model is rarely uncertain about which regime is active.

14 Results Comparison: Without HMM/GMM vs. With HMM/GMM

14.1 Combined Portfolio Metrics

Metric	Before	After	Change
Total P&L	₹26,572	₹38,975	**+₹12,403 (+47%)**
Total Return	$+5.31\%$	$+7.79\%$	**+2.48 pp**
Win Rate	62.7%	72.1%	**+9.4 pp**
Profit Factor	1.27	2.34	**+1.07 (+84%)**
Max Drawdown	-8.22%	-1.85%	**77% less risk**
Worst Trade	−₹7,692	−₹6,259	**Better loss control**
Number of Trades	67	43	Fewer, higher quality
Stop-Loss Exits	24	1	-96%

Table 20: Combined portfolio metrics, before vs. after.

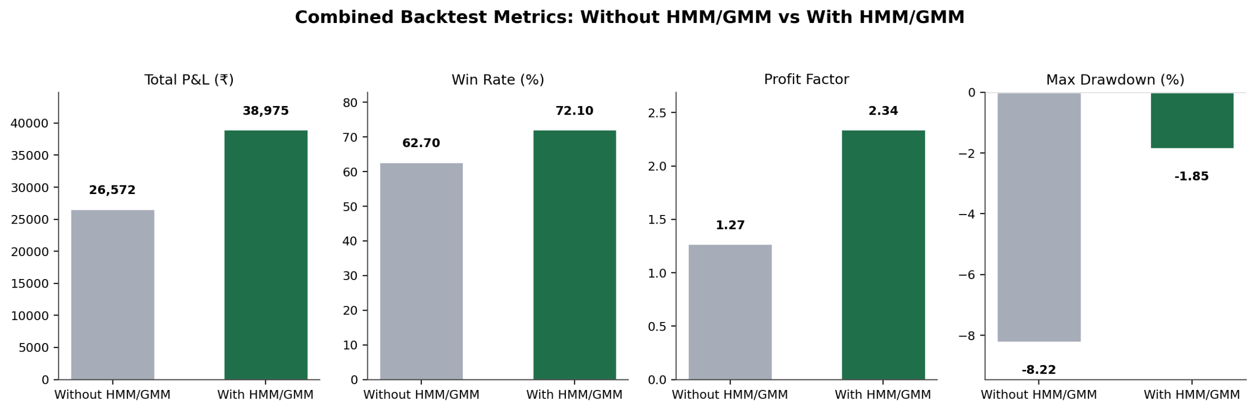

Figure 1: Combined portfolio metrics, before vs. after.

14.2 Strategy-Level Breakdown

Strategy	P&L Before	P&L After	Win% Before	Win% After	PF Before → After
Iron Condor	−₹16,806	**+₹2,714**	36.4%	55.6%	$0.61 \to 1.53$
Credit Put Spread	−₹12,983	−₹1,283	53.3%	60.0%	$0.59 \to 0.74$
Long Straddle	+₹20,123	+₹16,036	100%	83.3%	$\infty \to 3.56$
Bull Put Spread (Dip)	+₹36,238	+₹21,508	84.6%	78.3%	$2.52 \to 2.67$

Table 21: Strategy-level P&L, before vs. after regime gating.

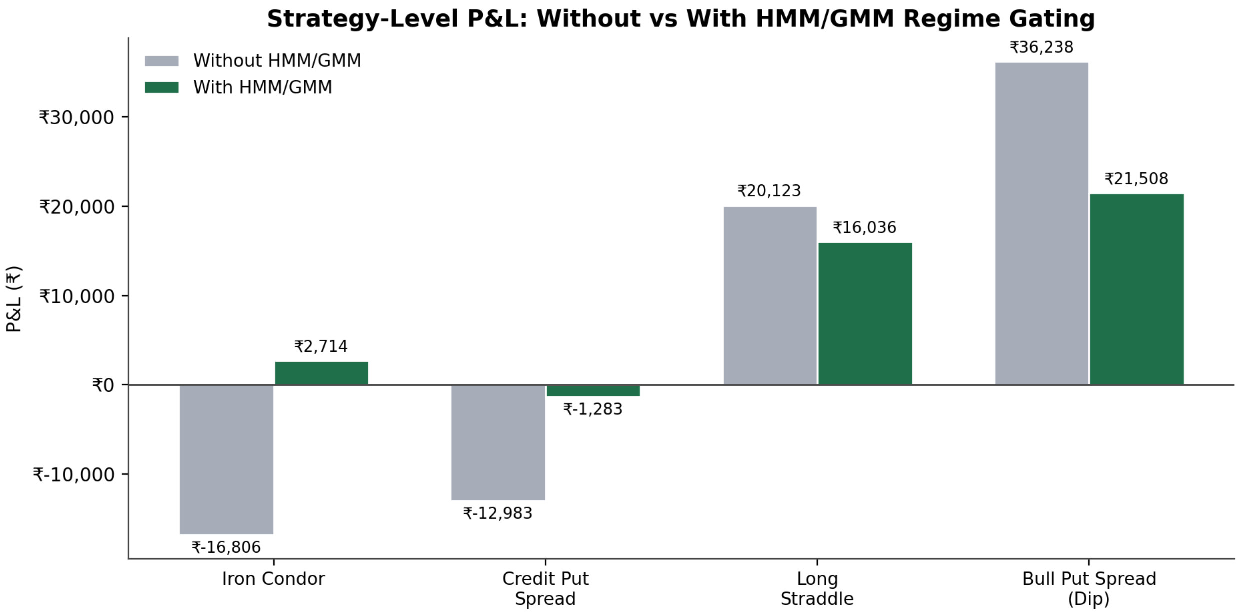

Figure 2: Strategy-level P&L, before vs. after regime gating.

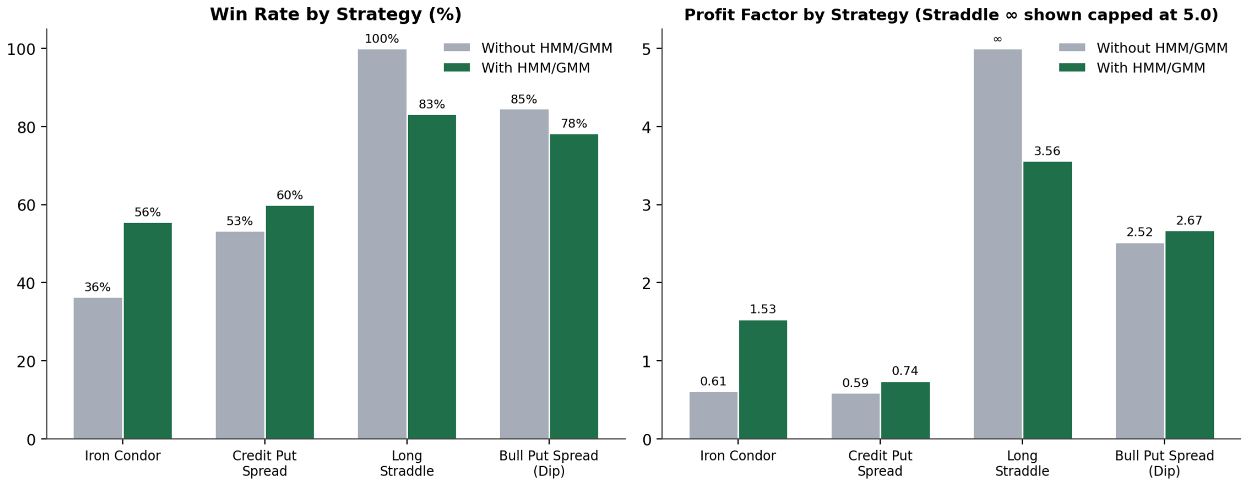

Figure 3: Win rate and profit factor by strategy, before vs. after.

Key Result

Iron Condor improved from −₹16,806 to **+₹2,714** — a ₹19,520 swing — because regime detection prevented 13 of 14 stop-loss entries. Only 9 trades were taken (vs. 22 before), but at much higher quality.

14.3 Exit Reason Distribution

The most telling change is not in the P&L itself, but in *why* trades closed. Without HMM/GMM, stop-losses accounted for 24 of 67 exits (36%). With HMM/GMM, only 1 of 43 exits (2%) was a stop-loss — most trades now reach their profit target, time exit, or the newly introduced regime-based exit.

Exit Reason	Before	After
PROFIT_TARGET	35	21
STOP_LOSS	24	**1**
TIME_EXIT	4	9
LEG_LOCK_PROFIT	4	5
REGIME_EXIT (new)	0	5
VEGA_EXIT (new)	0	1
IV_CRUSH_EXIT (new)	0	1

Table 22: Exit reason counts, before vs. after.

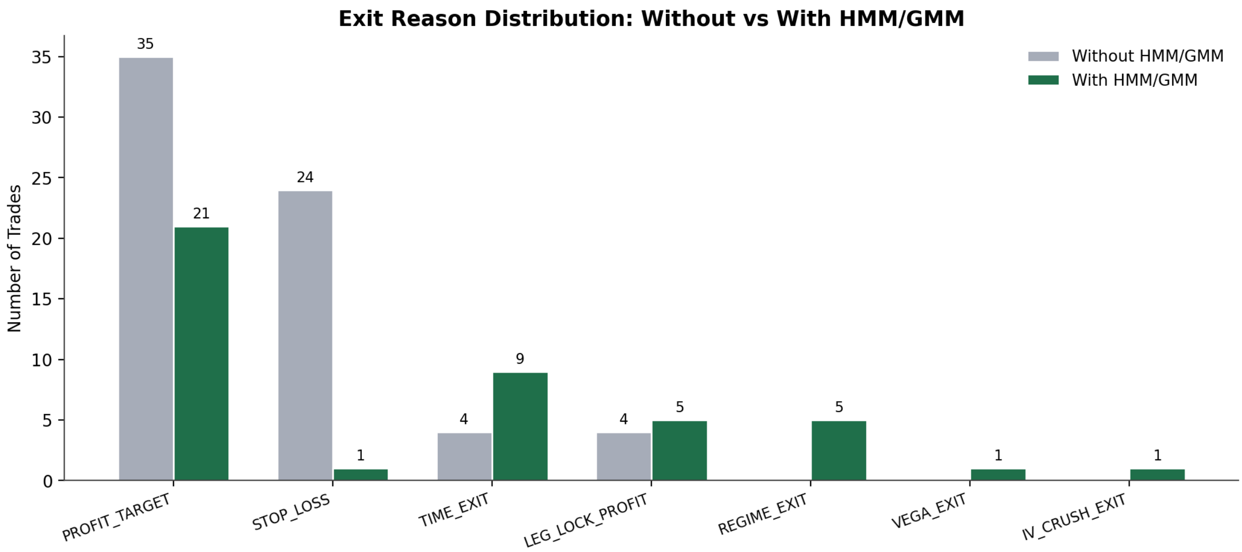

Figure 4: Exit reason counts, before vs. after.

14.4 Equity Curve (With HMM/GMM)

The cumulative P&L curve below is built directly from the HMM/GMM-enabled trade log (43 trades, 2022–2024). The early part of the curve is flat-to-slightly-negative while the walk-forward model is still building its first regime estimates on limited history; performance compounds more cleanly in the second half of the backtest once the regime classifier has more data to work with.

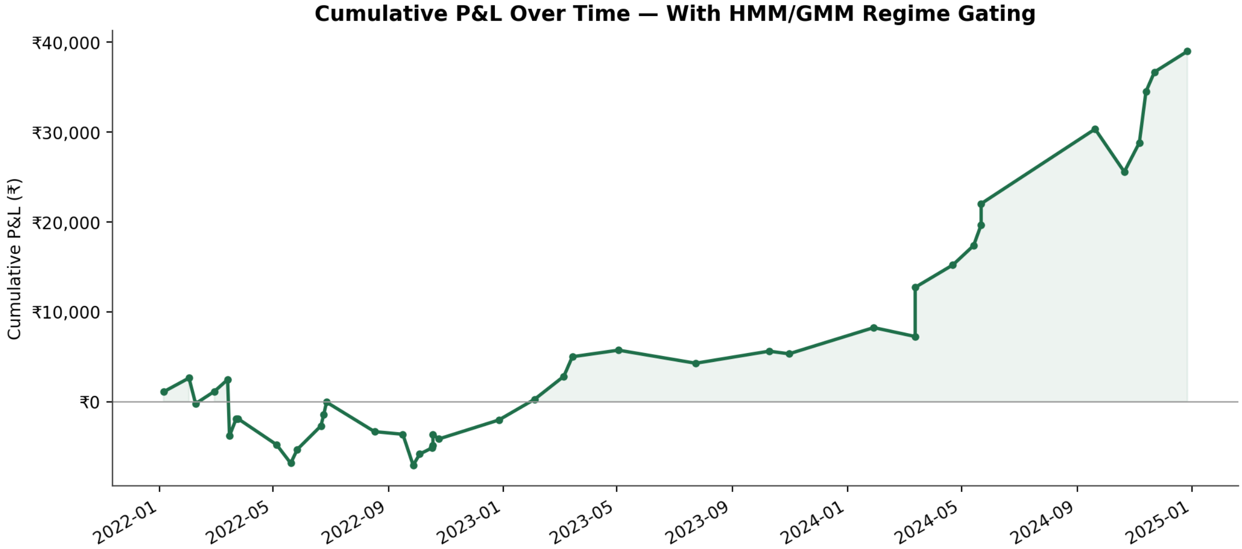

Figure 5: Cumulative P&L over the backtest window, with HMM/GMM.

15 Key Technical Details

- **Walk-forward training**: 200-day training window, re-fit every 63 trading days (quarterly) — 9 re-fit windows across the backtest, with no look-ahead bias.
- **Features used for regime detection**: log returns, 20-day historical volatility, IV/HV ratio, RSI(14), volume z-score, Bollinger Band width.
- **Regime construction**: GMM discovers statistical clusters in the feature space; HMM is layered on top to add temporal persistence (smoothing); resulting regimes are sorted by volatility (lowest = trending, highest = stress).
- **Average regime-classification confidence** across the backtest: 87.76%.
- **New `REGIME_EXIT` logic** closes a trade early if the market regime shifts mid-trade (used in 5 trades).
- **Dynamic stop-loss multipliers** scale with regime volatility — wider stops in volatile regimes, tighter stops in calm regimes.

16 Verification Checklist

✓ Full backtest ran successfully end-to-end (1 Jan 2022 – 31 Dec 2024).

✓ Walk-forward design confirmed to prevent look-ahead bias (first 200 days used for training only, no trades taken).

✓ All 4 strategies run correctly with regime gating active.

✓ Dashboard output verified: `multi_strategy_dashboard.html`.

✓ Results file verified: `multi_strategy_results.json`.

✓ Profit factor improved from 1.27 to 2.34 (+84%).

✓ Maximum drawdown reduced from -8.22% to -1.85% (77% less risk).

✓ Stop-loss hits reduced from 24 to 1 (-96%).

17 Conclusion

Adding HMM/GMM-based regime detection to the backtester produced a clear, across-the-board improvement: higher total P&L, a higher win rate, a much higher profit factor, and a sharply lower drawdown — achieved with fewer trades. The largest single contributor was the Iron Condor strategy, which went from the largest loss-maker to a net profit simply by being kept out of unfavourable regimes. The walk-forward design ensures these results are not inflated by look-ahead bias, and the new regime-based exit logic adds a further layer of risk control mid-trade.

> **Overall**: regime gating should be considered a validated improvement to the strategy engine and is recommended for continued use and further extension to additional strategies.

Overall Conclusion

Taken together, the two parts of this project tell a single coherent story. Part I established a free, fully self-contained, Greek-driven options backtesting framework for NIFTY 50 and validated four systematic strategies against three years of data, producing a positive but uneven result (₹26,572 net P&L, profit factor 1.27, two of four strategies net negative). Part II then layered walk-forward HMM/GMM market-regime detection on top of that same engine, gating each strategy to the regime it is statistically best suited for — without altering any strategy's core entry/exit rules.

Metric	v2.0 Baseline (Part I)	+HMM/GMM (Part II)	Change
Total P&L	₹26,572	₹38,975	**+47%**
Profit Factor	1.27	2.34	**+84%**
Max Drawdown	-8.22%	-1.85%	**77% less risk**
Trades Taken	67	43	Fewer, higher quality

Table 23: End-to-end improvement across the full project, baseline to final engine.

Bottom Line

Regime gating is a validated, low-risk-of-overfitting improvement: it reuses the same four strategies and the same Black-Scholes/Greek framework from Part I, adds no look-ahead information, and improves every headline metric simultaneously. Recommended next steps span both parts — transaction-cost modeling, weekly expiries, and real NSE data integration (Part I, Section 11), alongside extending regime gating to any future strategies (Part II, Section 6).